Current source contract for coset trialities

Human Compression project: L122 current recovery

This addendum governs the current reading of the three complete papers listed below. They recover the corrected principal contract of Leemans and Stokes, *Coset geometries with trialities and their reduced incidence graphs*, Acta Math. Univ. Comenianae 88(3) (2019), 911–916. The fixed source has six pages. The current machine-readable source contract preserves all 36 original dispositions, including background and conventions. Independent whole-original acceptance is a separate review decision; no source, conjecture or measurement credit is granted here.

Complete retained scientific paper	Pages
Coset trialities, chamber recognition and reduced relations	4
Explicit coset examples and their full triality centralizers	4
A coset geometry with a type cycle but no triality	3

The first two papers and their programs are copied byte for byte from the previously frozen positive packet. Its references to an unresolved bare existential Lemma 1 record the state of that edition. They are *superseded only on that point* by the present contract and the third paper, not silently edited or omitted.

The remaining general statement is now adjudicated. The new coset example satisfies every printed hypothesis of Lemma 1: an actual automorphism cycles the three defining subgroups, and every flag extends to a chamber. Nevertheless its full correlation group has no order-three type cycle. It has 612 objects of each type, full typed group of order 7344 and full correlation group of order 44064. It is connected and flag-transitive, but neither firm nor residually connected; the printed lemma assumes neither additional condition. Consequently the unrestricted existential implication and unqualified Method 1 are false. The valid sufficient condition is that the cube of the induced map fixes every defining coset, in particular $\sigma^3 = 1$. The analogous unqualified r-type implication already fails at $r = 3$.

All other routes remain complete and unchanged. The first paper proves the exact induced-action criterion, the complete rank-three recognition argument and lossless directed reconstruction. Its explicit examples correct the source's last-type resets and the unqualified cover/orientation claims; voltage loops and semiedges remain distinguished. The second paper proves the six specified finite Suzuki/projective examples, their actual trialities and full reduced groups, all 20 Suzuki absolute objects and their subgeometry, and the Mathieu construction. The named classical Suzuki model and Mathieu action/outer-automorphism facts remain explicit retained inputs. Current markings are not asserted to equal the unprovided historical choices.

Complete proof and program boundary. All eleven scientific pages above remain paid, as does this addendum. The entire old GAP certificate and seed file, and the entire new integer certificate and reflection/generator data, are required finite premises. Neither a short receipt nor the discovery automorphism search substitutes for these files. Optional controls, operational drivers and archived exploration are separately inventoried. The source supplies no executable listing; historical software and full external proof costs are unknown, not zero. The symmetric software policy applies to any later measurement. There is no new Lean, conjecture answer or

numerical factor. This is complete corrected source adjudication, not an assertion that every literal original theorem follows unchanged.

Attribution and review. The source construction and reduction are credited to Leemans–Stokes. Enumeration_volume authored the positive recovery, and rewrite_termination authored the new coset counterexample and this assembly. Root distinctly accepted both scientific components. The current assembly still requires its own review; the immutable component and review manifests record the precise roles.